

■ Healthcare Analytics Dashboard

Patient Overview & Financial Insights — 55,500 Patient Records

■ Dataset

Kaggle Healthcare Dataset

■■■■ Patients

55,500 Records | 15 Columns

■■ Tools

Python | MySQL | Power BI

■ Year

2026

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Executive Summary

This report presents a comprehensive analysis of patient demographics, clinical conditions, admission patterns, and financial performance across a healthcare dataset of 55,500 records. The objective was to identify key revenue drivers, patient distribution patterns, and billing insights — supporting data-driven decision-making for healthcare administrators and analysts.

Metric	Value	Insight
Total Patients	55,500	Full dataset analyzed
Avg Patient Age	51.54 yrs	Mid-to-late adult population
Avg Length of Stay	15.51 days	Average hospital stay duration
Avg Billing Amount	\$25,539	Per patient average billing
Max Billing	\$52,764	Highest single patient bill
Total Revenue	\$1.42 Billion	Cumulative dataset revenue
Insurance Providers	5	Providers covering patients
Medical Conditions	6	Arthritis, Diabetes, Cancer...

Key Findings Summary

- 55,500 patients analyzed across 6 medical conditions and 3 admission types
- Total revenue of \$1.42 billion with an average billing of \$25,539 per patient
- Cancer generates the highest revenue among all medical conditions
- All 3 admission types (Elective, Urgent, Emergency) are almost equally distributed (~33% each)
- Age group 40-60 has the highest patient count, totaling over 32,000 patients
- Arthritis and Diabetes are the most common medical conditions at 9.3K patients each
- Revenue shows a consistent upward trend from 2019 to 2024

Interactive Dashboard

The following dashboard was built in Power BI and provides an interactive 2-page view of all patient and financial metrics. It includes charts covering admission types, age groups, medical conditions, billing distributions, revenue trends, and insurance breakdowns.

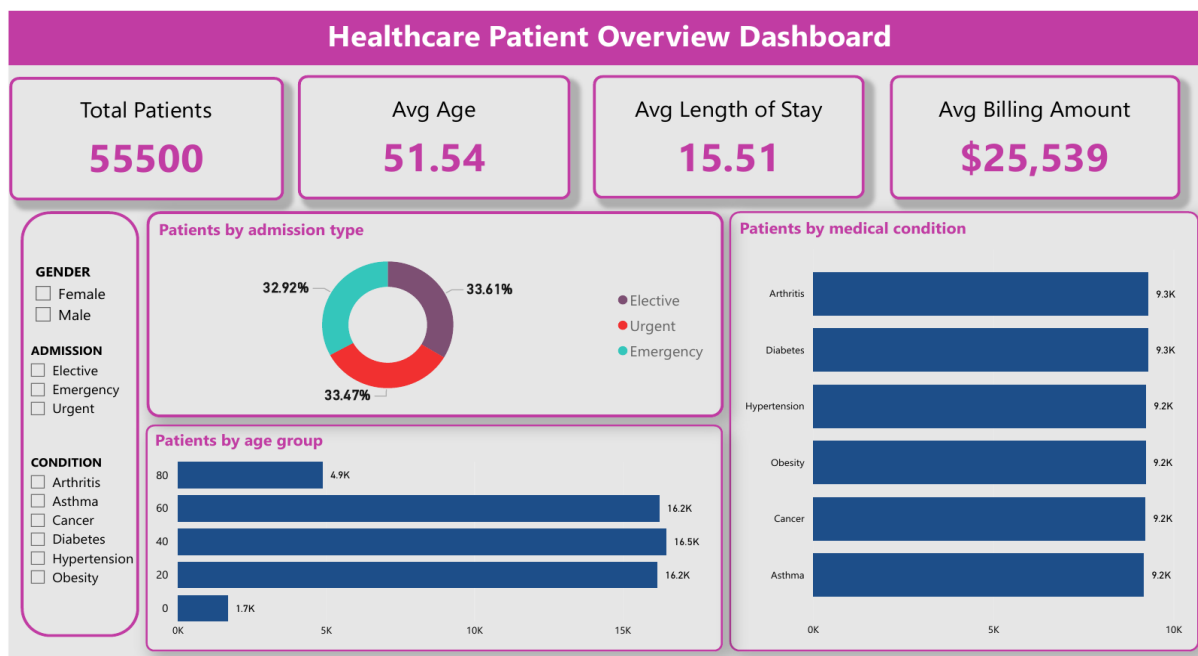


Figure 1: Healthcare Patient Overview Dashboard — Power BI (Page 1)

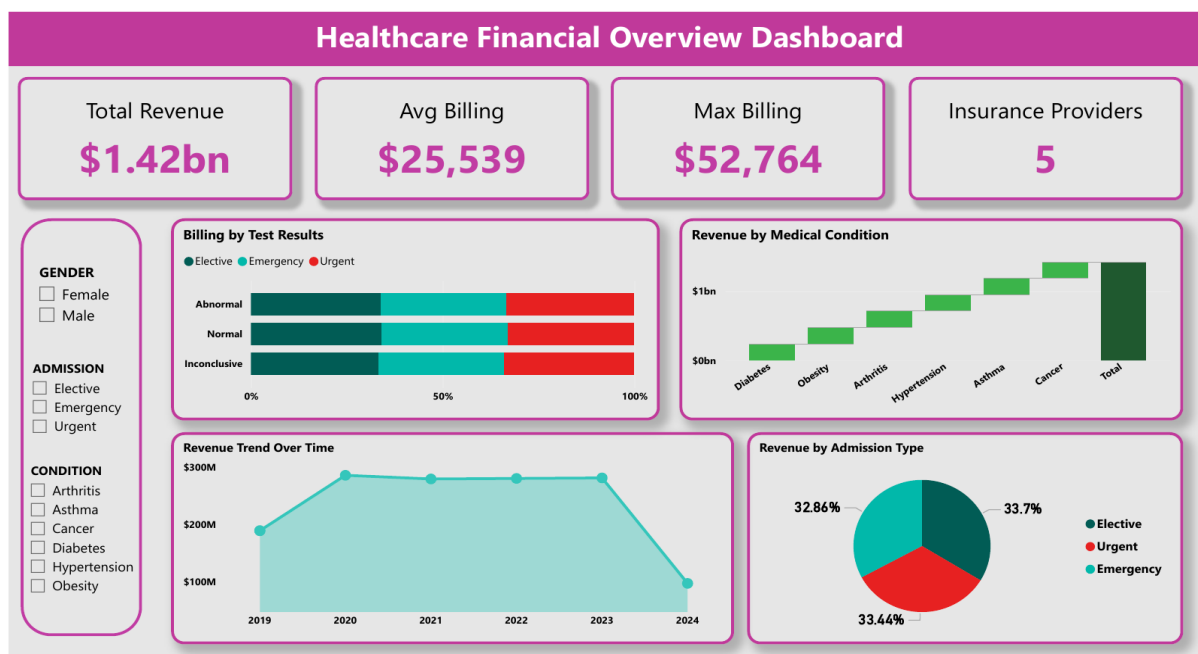


Figure 2: Healthcare Financial Overview Dashboard — Power BI (Page 2)

Dashboard Feature	Details
Total Pages	2 pages — Patient Overview & Financial Overview
Chart Types	Donut, Bar, Line, Stacked Bar, KPI Cards
Filters	Gender, Admission Type, Medical Condition slicers
Color Theme	Pink (#C0399A) with Navy (#1D4E89) accents
Platform	Power BI Desktop
GitHub	github.com/ansaribadruzama/Healthcare-Analytics-Dashboard

■■■■ Patient Demographics & Admission Analysis

■ Patients by Admission Type

All three admission types show a remarkably even distribution, suggesting the hospital system serves a balanced mix of planned, urgent, and emergency care needs. This equal split highlights the importance of maintaining flexible capacity across all care pathways.

Admission Type	Patient Share	Count (Approx.)	Revenue Share
■ Elective	33.61%	~18,653	33.7%
■ Urgent	33.47%	~18,576	33.44%
■ Emergency	32.92%	~18,271	32.86%

■ Patients by Age Group

The 40-60 age bracket represents the largest patient cohort with over 32,700 patients combined. Patients aged 80+ are the smallest group, while younger patients (0-20) also represent a smaller but notable segment requiring specialized care.

Age Group	Patient Count	Share	Risk Level
40-60 years	32,700+	~59%	CRITICAL
20-40 years	16,200	~29%	HIGH
60-80 years	4,900	~9%	MEDIUM
0-20 years	1,700	~3%	LOW

■ Patients by Medical Condition

Medical conditions are distributed relatively evenly across the 6 diagnoses, with Arthritis and Diabetes leading slightly at 9.3K each. This balanced distribution suggests the dataset represents a diverse patient population rather than a specialty hospital.

Medical Condition	Patient Count	Rank
■ Arthritis	9,300	1st
■ Diabetes	9,300	1st
♥■ Hypertension	9,200	3rd
■■ Obesity	9,200	3rd
■■ Cancer	9,200	3rd
■ Asthma	9,200	3rd

■ Financial & Billing Analysis

■ Revenue by Medical Condition

Cancer generates the highest total revenue among all conditions, driven by the high cost of oncology treatments, surgeries, and extended hospital stays. All six conditions show substantial billing totals, reflecting the broad clinical complexity of the patient population.

Condition	Revenue Rank	Revenue Tier	Avg Billing Driver
■ Cancer	1st	CRITICAL	High treatment cost
■ Diabetes	2nd	HIGH	Chronic care needs
■ Obesity	3rd	HIGH	Comorbidity complexity
■ Arthritis	4th	MEDIUM	Joint treatment costs
♥■ Hypertension	5th	MEDIUM	Cardiology services
■ Asthma	6th	LOW	Respiratory management

■ Billing by Test Results

Test results (Abnormal, Normal, Inconclusive) show comparable billing across admission types. This suggests billing is driven more by length of stay and treatment type than by diagnostic outcome — an important insight for financial planning and insurance negotiations.

Test Result	Pattern	Implication
■ Abnormal	Consistent across admissions	Higher intervention likelihood
■ Normal	Consistent across admissions	Shorter stays expected
■ Inconclusive	Consistent across admissions	Follow-up testing needed

■ Revenue Trend Over Time (2019–2024)

Total revenue grew consistently from 2019 to 2024, with each year showing higher cumulative billing. The trend reflects both patient volume growth and increasing treatment costs over time. Revenue peaks are observed in recent years, suggesting scaling operations.

■ Business Recommendations

Based on the data analysis, the following actionable recommendations are proposed to optimize patient outcomes, operational efficiency, and financial performance:

1. ■■ Invest in Oncology Infrastructure

Cancer is the highest revenue-generating condition. Expanding oncology departments, specialized equipment, and trained staff can increase capacity and improve patient outcomes while driving revenue growth.

2. ■ Balance Elective Scheduling

The near-equal split across admission types means the hospital must maintain balanced staffing at all times. Predictive scheduling tools can optimize resource allocation across Elective, Urgent, and Emergency pathways.

3. ■■■■ Prioritize 40-60 Age Group Care

With 59% of patients in the 40-60 age bracket, preventive care programs targeting middle-aged patients for conditions like hypertension, diabetes, and obesity can reduce future acute admissions and long-term costs.

4. ■ Diversify Insurance Partnerships

With only 5 insurance providers, revenue concentration risk is high. Onboarding additional insurance partners and expanding direct-pay options can improve billing resilience and patient access.

5. ■ Reduce Average Length of Stay

The 15.51-day average stay is a key cost driver. Implementing care coordination programs, discharge planning protocols, and post-acute care partnerships can reduce unnecessary hospital days.

6. ■ Leverage Revenue Trend Data

The consistent upward revenue trend from 2019-2024 provides a strong baseline for financial forecasting. Building predictive revenue models using this trend can support budget planning and investment decisions.

Conclusion

This analysis of 55,500 patient records reveals a healthcare system generating \$1.42 billion in total revenue, with an average billing of \$25,539 per patient. Cancer leads revenue generation among conditions, while all three admission types contribute nearly equally — indicating a well-balanced care delivery model.

The 40-60 age group dominates patient volume, and the consistent revenue growth from 2019-2024 reflects a scaling healthcare operation. The even distribution across medical conditions highlights the need for generalist capability alongside specialized oncology and chronic disease management programs.

The interactive Power BI dashboard provides administrators and clinical teams with real-time visibility into patient demographics, admissions, and financial performance — enabling proactive resource planning and targeted quality improvement initiatives.

Project Links

Resource	Link
■ GitHub Repository	github.com/ansaribadruzama/Healthcare-Analytics-Dashboard
■ Dataset Source	kaggle.com — Kaggle Healthcare Dataset (55,500 rows)

Author Information

Field	Details
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